Yoojung Han

Ph.D. Candidate

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EDUCATION

2020.09 – 2027.08 (Expected)	**Integrated M.S. & Ph.D.** Interdisciplinary Program in Bioinformatics Seoul National University, Korea
2016.03 – 2020.08	**B.S. (Cum Laude)** Department of Biosystems & Biomaterials Science and Engineering Seoul National University, Korea

PUBLICATIONS

([1]Marked authors contributed equally.)

3. **Targeted long-read sequencing for high-resolution repeat profiling in myotonic dystrophy type 1**
   Han, Y., Jang, J.-H., and Chang, H. (2026)
   Experimental & Molecular Medicine (2025 IF: 17.5)

   Analyzed clinical DNA nanopore sequencing data to profile extended repeat lengths and DNA methylation patterns, revealing patient-specific genomic heterogeneity in myotonic dystrophy type 1.

2. **VaxLab: integrated platform for rapid multistrategy mRNA vaccine design**
   Kim, J.[1], **Han, Y.[1]**, Kwon, C. Y., and Chang, H. (2026)
   Experimental & Molecular Medicine (2025 IF: 17.5)

   Engineered an automated pipeline that serves as a computational platform integrating diverse sequence optimization tools to advance the development of mRNA therapeutics.

1. **Another common genetic ataxia in South Korea: Spinocerebellar ataxia 36**
   Ahn, J. H., Lee, S., Moon, J., **Han, Y.**, Chang, H., Youn, J., Cho, J. W., and Jang, J.-H. (2025)
   European Journal of Human Genetics (2025 IF: 4.6)

   Contributed to the genetic characterization of spinocerebellar ataxia 36 by analyzing nanopore sequencing data to accurately measure allele-specific pathogenic repeat lengths.

MANUSCRIPTS IN PREPARATION

([1]Marked authors contributed equally.)

1. **Massively parallel screening of codon optimization strategies reveals context-dependent design rules for mRNA therapeutics**
   Han, Y.[1], Lee, S.[1], et al. (in preparation)

   A high-throughput MPRA study investigating diverse codon optimization variants across various endogenous genes, revealing that protein-specific factors dictate intracellular mRNA stability, thereby establishing novel design principles for advanced mRNA therapeutics including vaccines and protein replacement therapies.

2. **High-resolution profiling of TERRA alterations following G-quadruplex stabilization via nanopore direct RNA sequencing**
   Han, Y.[1], Ju, S.[1], et al. (in preparation)

   This study utilizes nanopore direct RNA sequencing to comprehensively profile telomere repeat-containing RNA (TERRA) in U2OS cells, identifying chromosome-specific changes in UUAGGG repeat lengths, RNA methylation, and poly(A) tail dynamics induced by G-quadruplex stabilizer.

3. **Direct nanopore single-molecule sequencing of DNA-RNA hybrids**
   Han, Y.[1], Lee, S.[1], et al. (in preparation)

   Developing a custom basecaller model for the nanopore direct RNA sequencing platform to decode DNA sequences, aiming to achieve single-molecule resolution of DNA-RNA hybrids and facilitate DNA barcode DRS multiplexing.

AWARDS AND GRANTS

2026.09	**ECCB Travel Fellowship Award** European Conference on Computational Biology Association
2026.02	**Young Scientist Award** Korea Genome Organization
2025.09 – 2027.08	**Next Generation Scholarship** National Research Foundation of Korea ($17,000 / year)
2024.03 – 2027.02	**Next Generation Scholarship in Basic Studies** Seoul National University ($16,500 / year)
2024.10	**Best Poster Award** Korean Society for Bioinformatics

CONFERENCE PRESENTATIONS

2026.09 (Upcoming)	**Poster | European Conference on Computational Biology** Geneva, Switzerland
2026.07	**5-min Talk | Genomics and Computational Biology Symposium, KSMCB** Gyeongju, Korea
2026.02	**Oral | Korea Genome Organization Winter Symposium** Hongcheon, Korea
2024.11	**Poster & 1-min Talk | Cold Spring Harbor Laboratory Meeting (Biological Data Science)** New York, USA
2024.10	**Poster | International Conference of Korean Society for Bioinformatics** Gyeongju, Korea
2023.11	**Poster | International Conference of Korean Society for Bioinformatics** Yeosu, Korea
2022.10	**Poster | International Conference of Korean Society for Bioinformatics** Daejeon, Korea

PATENTS

2023.12	**Method for providing information for the diagnosis of ataxia** Jang, J.-H., Cho, J. W., Ahn, J. H., Chang, H., and **Han, Y.** *Korean Patent Application No. 10-2023-0197358* (Pending)
2023.10	**Method for providing information for diagnosing repeat expansion disorder diseases using the CRISPR-Cas9 system and nanopore sequencing analysis** Jang, J.-H., Chang, H., **Han, Y.**, Cha, J., and Park, J.-H. *Korean Patent Application No. 10-2023-0147875* (Pending)

SKILLS

Computer languages	Python, R, Bash/Shell scripting
Machine learning	scikit-learn, PyTorch, TensorFlow
Infrastructure	SLURM, Snakemake
Wet lab experiments	Mammalian cell culture & transfection, molecular cloning, *in vitro* transcription (IVT), and advanced RNA handling & QC
Sequencing technologies	**Nanopore (DNA/RNA):** End-to-end workflows (library prep to data analysis), sequence & modification profiling, raw signal processing, and custom basecaller model development & fine-tuning. **Illumina:** RNA-seq library preparation and transcriptomic data analysis.